Interpreting Results & Crafting the Narrative

ECON 692 – Applied Economics Seminar

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Today's Agenda

Time	Duration	Activity
4:35	45 min	Guest speaker: USF Career Services
5:20	10 min	Break
5:30	30 min	Interpreting results & the empirical narrative
6:00	30 min	Claude Code: slash commands & adversarial agents
6:30	60 min	Work sprint: revise results + adversarial review
7:30	25 min	Share-outs: what did your reviewer catch?
7:55	20 min	Wrap-up & next steps

Interpreting Results

What Goes in a Results Section?

Three jobs:

1. **State the main finding.** Put it in the first sentence.
2. **Unpack it.** Sign, magnitude, significance, and what they mean economically.
3. **Stress-test it.** Robustness, alternative specs, threats.

Your reader should know your main result by the end of the first paragraph.

Interpreting Coefficients

For every coefficient you report, can you answer:

- **Sign:** Does the direction make sense?
- **Magnitude:** Is the size plausible?
- **Statistical significance:** Significant at what level?
- **Economic significance:** Does the effect size *matter*?
- **Units:** Can you say it in plain English? ("A 1 SD increase in X is associated with a $Y change in Z.")

If you can't answer all five, keep working.

Worked Example

DiD of a job training program on wages. Training $\times$ Post $= 1{,}240$** (SE $= 480$).

- **Sign:** Positive. Training raises wages. Makes sense.
- **Magnitude:** $1,240/yr $\approx$ $103/month. Plausible for a short program.
- **Stat. sig.:** $p < 0.05$ without controls, $p < 0.10$ with controls. Sensitive.
- **Econ. sig.:** 4% of a $30K base salary. Modest but real.
- **Plain English:** “Participants earned about $1,200 more per year after training.”

With controls the estimate drops to 890* (SE $= 510$). That sensitivity matters and should be discussed.

Null Results

Some of you got null results. That's fine, and they can be informative.

Do:

- Report them honestly
- Discuss *why* the effect might be zero
- Frame as a contribution: "*X* doesn't survive [your check]"

Don't:

- Say "insignificant" and move on
- Keep running specs until something lights up
- Treat $p = 0.06$ and $p = 0.04$ as fundamentally different

A well-discussed null is worth more than an unexplained significant result (Gelman & Stern, 2006).

Common Mistakes

Things to watch out for:

- **No regression table.** Descriptive figures are great, but show me at least one coefficient with standard errors.
- **Ignoring red flags.** If your placebo test fails or your R^2 is negative, *talk about it*. Don't hope nobody notices.
- **Over-claiming.** "Confirms a causal effect" when your CI includes zero? No, it doesn't.
- **Missing code.** If I can't see how you got your numbers, I can't evaluate them.
- **Mislabeled figures.** Check your axis labels. Then check them again.

Structuring the Narrative

Your results section tells a story. Each step builds credibility:

Main spec	→	Robustness	→	Heterogeneity	→	Threats
Your preferred model, fully interpreted		Alternative specs that support or challenge		Subgroups, time periods, context		Reverse causality, OVB, measurement

$\Downarrow$

Bottom line: one paragraph. What do we learn?

Claude Code: Slash Commands & Adversarial Agents

Why Use AI as a Reviewer?

When you've been staring at your own analysis for weeks, you lose perspective. You stop noticing gaps, you assume the reader has context they don't, and you skip over weak spots.

Setting up an adversarial reviewer in Claude Code gives you a way to get a fresh read on your work whenever you want one.

Slash Commands

A slash command is just a reusable prompt saved as a file.

Create `.claude/commands/review-results.md` in your project:

```
# .claude/commands/review-results.md

Read my results section and check:
1. Every coefficient has sign, magnitude,
   significance, and economic interpretation
2. Figures have labeled axes and titles
3. Robustness checks are present
4. Limitations are acknowledged
Flag anything missing or unclear.
```

Type `/review-results` in Claude Code. Done.

Commands You Should Build

Command	**What it does**
`/review-results`	Checks coefficient interpretation
`/check-tables`	Audits figure/table labels
`/check-robustness`	Lists threats you haven't addressed
`/summarize-findings`	Writes a 1-paragraph summary
`/devil`	Plays devil's advocate on your ID strategy

Each one is a markdown file in `.claude/commands/`. Make as many as you want.

My Slash Commands: Paper Reviewers

Each one acts as a different type of referee:

Command	Role
`/content-editor`	Journal editor: argument flow, contribution, consistency
`/econometrics-editor`	Econometrician: ID strategy, specs, interpretation
`/style-editor`	Style editor: flags AI-sounding prose, tightens writing
`/rct-editor`	Field experimentalist: RCT design, compliance, attrition
`/latex-copy-editor`	Copy editor: grammar, citations, LaTeX formatting

Why specific commands? Each has a focused role with clear instructions. A targeted reviewer with domain expertise gives better feedback than a generic "review my paper" prompt.

My Slash Commands: Research Tools

Command	What it does
`/review-r`	R code quality, reproducibility, domain correctness
`/data-analysis`	End-to-end analysis: exploration to tables
`/lit-review`	Literature search, synthesis, gap identification
`/interview-me`	Turns a vague idea into a research design
`/devils-advocate`	Challenges your design with pointed questions

You can build your own for your capstone. Think about what kind of feedback you need most and write a prompt for it.

The Adversarial Reviewer: `/devil`

Here's an example:

```
# .claude/commands/devil.md

You are an economics professor grading a
masters-level capstone. Read my empirical
strategy and results, then:

1. What is the biggest threat to my causal
   claim? Be specific.
2. What robustness check would most weaken
   my results if it failed?
3. What alternative explanation haven't I
   considered?
4. If you had to reject this paper, what
   would you say?

Be direct. Don't be nice.
```

CLAUDE.md Context Matters

The reviewer is only as good as the context you give it.

Your `CLAUDE.md` should include:

- **Research question** (one sentence)
- **ID strategy** (what variation are you exploiting?)
- **Data** (unit of observation, time period, sample size)
- **Key variables** (treatment, outcome, controls)
- **Known threats** (what are you worried about?)

Pull up your CLAUDE.md right now. Does it have all five?

Example CLAUDE.md

```
# My Capstone Project

## Research Question
Does social media sentiment cause stock
price movements?

## Identification Strategy
Firm-level FE + DoubleML with XGBoost.
Treatment: daily Twitter sentiment score.

## Key Concern
Reverse causality -- prices may drive
sentiment, not the other way around.
Placebo test (current sentiment predicting
lagged returns) is a critical check.

## Data
Bloomberg daily panel, 160K obs, 2018-2025.
```

The Feedback Loop

1. Write or revise your results section
2. Run `/devil`. Read the critique.
3. Fix what you agree with. Push back on what you don't.
4. Run `/review-results` to check formatting and completeness
5. Repeat before your next submission

The goal is catching blind spots before I do.

Work Sprint

Work Sprint (6:30–7:30)

Two tasks:

1. **Work on your results.** Write, revise, improve.
2. **Set up and run an adversarial review.**
   - Create `.claude/commands/devil.md`
   - Make sure your `CLAUDE.md` has enough context
   - Run `/devil` and see what it catches

I'll come around for check-ins.

Share-outs & Wrap-up

Share-outs (7:30–7:55)

What did your adversarial reviewer catch?

Quick round, 2 minutes each:

- What was the most useful critique?
- Did it find something you hadn't thought of?
- What are you going to change?

Next Week & Deadlines

Next week (4/9): Writing the intro, background, and empirical strategy sections.

Upcoming:

- Weekly progress report due **Wednesday**
- **Full Draft due April 24** (three weeks)

Between Now and the Full Draft

You need a complete paper: intro, background, empirical strategy, results, conclusion. Start writing prose this week. Your results section should pass the five-question test by next week.